

Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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Summary

Data Scientist & Machine Learning Engineer with strong leadership in machine learning and statistical modeling who generated over \$1.3 billion in business value. Specializations include design of data solutions, implementation of MLOps pipelines on AWS and Snowflake, enhancing customer engagement.

Education

Bachelor of Science in Mechanical Engineering - Arizona State University - Tempe, AZ 06/2003 to 05/2006
STEM, Multivariate Calculus, Differential Equations, Linear Algebra, Numerical Methods, Algorithms, Data Structures, Statistics, Mechanics, Material Science, Manufacturing, Finite Element Analysis, Computational Fluid Dynamics, Internal Combustion Engines, Mechanisms, Heat Transfer, Thermodynamics, Manufacturing, Economics, Cost, Finance, Control Systems, Design

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **US Department of Labor:** Operating Engineer Vocational Certificate

Professional Experience

Lead Data Scientist - Caterpillar Inc. - Surprise, AZ

09/2022 to Present

- Architected and developed end-to-end data products using programming skills in Python and R, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Used LLMs and natural language processing to find similarities in service jobs and performed feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Trained AI models using Python and SageMaker, applying machine learning algorithms like neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and enhanced the accuracy of equipment repair time predictions, leading to better maintenance scheduling
- Engineered production-ready MLOps pipelines on AWS and Snowflake, establishing best practices for cloud architecture. Managed project lifecycles, automated enterprise networks, and implemented monitoring strategies to assess model performance and business outcomes.

Reliability Analytics Manager - Amazon.com - Tempe, AZ

02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis, and wrote a white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Utilized advanced data science techniques to extract actionable insights from research to driving strategic decision-making and improving operational outcomes
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager - Empire-Cat - Mesa, AZ

01/2021 to 02/2022

- Developed a time-series MLOps pipeline for AI-driven forecasting in supply chain management, integrated data using SQL, R and Python, managed a component exchange program, and collaborated with clients on durability forecasting, enhancing supply chain efficiency
- Leader of Empire's mining machine component exchange program, focused on supply and demand of diesel engines, transmission, torque converters, etc...
- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days.

Reliability Engineering Manager - Empire-Cat - Mesa, AZ

01/2016 to 01/2021

- Supervised 8 engineers and technical communicators and acted as a customer-facing expert on mining equipment. Mentored up to 6 interns per year
- Headed up the root cause failure analysis program for Empire by taking a data, facts, and evidence approach, which fed the FMEA and 6 Sigma process at Caterpillar
- Implemented a data-driven failure analysis program, boosting an aging fleet's physical availability to over 90%

Technical Communications Manager - Empire-Cat - Mesa, AZ

12/2012 to 01/2016

- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding \$1.2 billion in savings over 6 years
- Client-facing role to communicate voice of the customer to Caterpillar's engineering organizations by leading a team of 5 Technical Communicators
- Leader with business acumen to build trust and enhance customer experience and relationships to promote data-based decisions
- Partnered on design of experiments to improve reliability, safety and durability.
- Developed Python and MS-SQL application to track performance and outcomes of prototype designs tested in the field.

Reliability Engineer - Empire-Cat - Mesa, AZ

01/2012 to 12/2012

- Created statistical oil analysis application to analyze heavy equipment fleets according to ASTM D 7720 and condition monitoring practices
- Served as subject matter expert for mechanical engineering and heavy equipment.
- Researched top reasons causing downtime for machinery operating at mine sites

Technical Communicator, Technical Consultant - Road Machinery, LLC - Phoenix, AZ 01/2008 to 01/2012

- Delivered customer-facing engineering support reliability engineering from a facts, data and evidence perspective
- Invoiced \$2.5M in consultancy fees over 3-½ years by providing specialized consultancy of mining fleet, and life cycle management
- Mentored a team of engineers at a gold mine in central Mexico, enhancing their technical skills
- Developed total cost of ownership forecast for 18 haul trucks using MATLAB and statistical models, which resulted in a sale of \$72M electric haul trucks
- Conducted investigations and authored root cause failure analysis reports for high-horsepower diesel engines, transmissions and other off-highway earth moving equipment
- Designed tools and below-the-hook lifting devices to enhance workplace safety, compliance and ease of use

Design Engineer - Komatsu North America - Chattanooga, TN 06/2006 to 12/2007

- Designed and refined mechanical systems and components, improving product durability through FRACAS
- Contributed to prototyping and testing, analyzing performance data, and implemented product improvements and corrective actions
- Prepared comprehensive technical documentation and communicated project updates, improving team alignment and ensuring timely completion of deliverables.
- Leveraged Pro Engineer and finite element analysis to optimize engineering designs

Engineering Intern - Xtek Mining Services - Tempe, AZ 01/2005 to 09/2005

- Exhibited competence in data visualization, analysis, advanced Excel, VBA, Access, and HTML
- Conducted inspections and reverse engineering of large gears and shafts for mining shovels and draglines
- Reverse-engineered gear geometry data and pioneered an change gear optimization application to minimize manufacturing defects
- Drafted and detailed engineering drawings, enhancing the clarity and accuracy of design documentation

Operating Engineer - Ryan Central Inc. - McHenry County, IL 06/2002 to 09/2002

- Operated heavy equipment a mass-dirt excavation sites, including the development of roads, building pads, lakes on construction projects
- Loaded scrapers in push-pull configuration

Heavy Equipment Mechanic - Roland Machinery Co. - Marengo, IL 04/2000 to 06/2002

- Developed strong troubleshooting skills, diagnosing and resolving mechanical issues promptly
- Conducted maintenance and repair of heavy equipment, diesel engines, and hydraulic systems to ensure optimal performance
- Skillfully repaired Komatsu machinery and Cummins diesel engines, guaranteeing peak operational efficiency
- Executed welding repairs on booms, structures and buckets
- Skillfully re-manufactured key components including engines, transmissions, final drives, and hydraulic cylinders
- Proven capability in handling Class B CDL service vehicles, truck-mounted cranes, and overhead cranes, with a steadfast commitment to safety and efficiency in all operations

Heavy Equipment Apprentice Mechanic - Harry W. Kuhn - West Chicago, IL 12/1997 to 04/2000

- Serviced various heavy machinery such as Caterpillar, Komatsu, Bomag, Ingersoll-Rand, and Gradall, along with proficiency in repairing diesel engines

- Developed expertise in the safe loading and unloading of heavy machinery on transport trailers, ensuring secure transit and reliable operations

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn

Courses

- **Amazon Machine Learning University:** Tabular Data with Autogluon
- **ASME:** Y14.5M Geometric Dimensioning and Tolerancing
- **Caterpillar:** Applied Failure Analysis • SQL Server Toolset & Services • Business Tools for Service Managers • Improving Component Durability
- **Data Camp:** Bayesian Modeling in RJAGS
- **DDI:** Communicating for Leadership Success • Coaching for Peak Performance • Delegating with Purpose • Setting Performance Expectations • Resolving Workplace Conflict
- **DeepLearning.AI:** Building Systems with the ChatGPT API • ChatGPT Prompt Engineering for Developers • Finetuning Large Language Models
- **Franklin Covey:** Leadership: Great leaders - Great Teams - Great Results
- **Harper College:** Pneumatics and Hydraulics
- **IBM:** Advanced RAG with Vector Databases and Retrievers • Agentic AI with LangChain and LangGraph • Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI • Build RAG Applications • Deep Learning with Keras and Tensorflow • Deep Learning & Neural Networks with Keras • Vector Databases for RAG
- **LinkedIn Learning:** Advanced Snowflake: Deep Dive Cloud Data Warehousing and Analytics • Building Applications Using Amazon Bedrock • Data Engineering with dbt • Designing Machine Learning Systems in the Cloud • Learning Kubernetes
- **Noria:** Oil Analysis I, II, III
- **O'Reilly Media:** Designing Machine Learning Systems in the Cloud
- **SAE:** Principles of Cost and Finance for Engineers